

Collection of Linear Algebra Problems

Zeta Zero

For corrections or to suggest a new problem for this collection, please email `category.theory.for.cats@gmail.com`.

1. (Vandermonde determinant) Let

$$V = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Show that $\det V = \prod_{1 \leq i < j \leq n} (x_j - x_i)$.

2. (Putnam 2008/B6) Let n and k be positive integers. Say that a permutation σ of $\{1, 2, \dots, n\}$ is k -limited if $|\sigma(i) - i| \leq k$ for all i . Prove that the number of k -limited permutations of $\{1, 2, \dots, n\}$ is odd if and only if $n \equiv 0$ or $1 \pmod{2k+1}$.
3. Let A and B be $n \times n$ matrices satisfying $A + B = AB$. Show that $AB = BA$.
4. (Putnam 1986/B6) Suppose A, B, C, D are $n \times n$ matrices, satisfying the conditions that AB^T and CD^T are symmetric and $AD^T - BC^T = I$. Prove that $A^T D - C^T B = I$.
5. Do there exist square matrices A and B such that $AB - BA = I$?
6. (Putnam 2003/B1) Do there exist polynomials $a(x), b(x), c(y), d(y)$ such that $1 + xy + x^2 y^2 = a(x)c(y) + b(x)d(y)$ holds identically?
7. Let $A = (a_{ij})$ be a real $n \times n$ matrix satisfying $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all $1 \leq i \leq n$. Prove that A is invertible.

8. Let

$$A = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 2 \\ 1 & -2 & 0 \end{pmatrix}$$

and let I_3 denote the 3×3 identity matrix.

- (a) Verify that $A + I_3$ is nonsingular.
- (b) Verify that $(I_3 - A)(I_3 + A)^{-1} = (I_3 + A)^{-1}(I_3 - A)$.
- (c) Verify that the matrix $Q = (I_3 - A)(I_3 + A)^{-1}$ is orthogonal (i.e., $Q^T = Q^{-1}$).
- (d) Determine all real eigenvalues of the matrix Q , along with their corresponding eigenvectors.
- (e) Prove properties (a), (b), and (c) for the general case where A is an arbitrary 3×3 skew-symmetric matrix ($A^T = -A$).

9. Consider the quadratic form $q(x) = x^T Ax$, where

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix},$$

and $x \in \mathbb{R}^3$ is a non-zero column vector.

- (a) Determine the orthogonal transformation $x = Qy$ that reduces the quadratic form to its canonical form.
- (b) Find the extreme values of the function $g : \mathbb{R}^3 \setminus \{(0, 0, 0)^T\} \rightarrow \mathbb{R}$ given by $g(x) = \frac{x^T Ax}{x^T x}$, as well as the vectors where these values are attained.

10. Let $A \in M_2(\mathbb{C})$. Prove that for any positive integer $p \in \mathbb{N}^*$,

$$\det(A^p + I_2) = (\det A)^p + \operatorname{tr}(A^p) + 1.$$

11. Consider a matrix $A \in M_2(\mathbb{C})$ such that $\det A = 1$. Prove that

$$\det(A^2 + I_2) + \det(A^2 + 2A - I_2) = 8.$$

12. Let $A \in M_2(\mathbb{R})$ such that $\det A = d \neq 0$ and $\det(A + dA^*) = 0$, where A^* denotes the adjugate matrix of A . Prove that $\det(A - dA^*) = 4$.

13. Let $x > 0$ be a real number and let $A \in M_2(\mathbb{R})$ be a matrix such that $\det(A^2 + xI_2) = 0$. Prove that $\det(A^2 + A + xI_2) = x$.

14. Let $A \in M_2(\mathbb{R})$. Prove that

$$\det(A^2 + A + I_2) \geq \frac{3}{4}(1 - \det A)^2.$$

15. Let $a, b \in \mathbb{R}$ such that $b - a^2 > 0$. Find all matrices $A \in M_2(\mathbb{R})$ satisfying $\det(A^2 - 2aA + bI_2) = 0$.

16. Let $A \in M_3(\mathbb{C})$ be a matrix such that all entries on the main diagonals of A and A^2 are zero. Prove that if $|\det A| < 1$, then $\det(I_3 + A^n) \neq 0$ for all $n \in \mathbb{N}^*$.

17. Let $A \in M_3(\mathbb{C})$ be an invertible matrix. Prove that if $A^T A^{-1} = B$, then

$$\det(I_3 + B) = 2(1 + \operatorname{tr} B) \quad \text{and} \quad \det(I_3 + B^2) = 2(1 + \operatorname{tr} B^2).$$

18. Let $A, B \in M_3(\mathbb{C})$ such that $(AB - BA)(AB - BA - I_3) = O_3$. Prove that $AB = BA$.

19. Let $A \in M_4(\mathbb{R})$ be a symmetric matrix. Prove that $\det(A + I_4) = 4\sqrt{\det(A^2 + I_4)}$ if and only if $A = I_4$.

20. Let $A \in M_n(\mathbb{R})$. If $AA^T = I_n$, prove that $|\operatorname{tr} A| \leq n$.

21. Let $n \geq 2$ be an integer, and let $A, B, C \in M_n(\mathbb{C})$ be three mutually commuting matrices such that $A^{1999} = B^{1999} = C^{1999} = O_n$. Prove that the matrix $I_n + A^2 + B^2 + C^2$ is invertible.

22. Let $A \in M_n(\mathbb{Q})$ be a matrix such that $\det(A + \sqrt[n]{2}I_n) = 0$. Prove that

$$\det(A - I_n) = \det A + (\det(A + I_n))^n.$$

23. Let $A, B \in M_n(\mathbb{C})$ be two matrices such that $A^2 - B^2 = 2(AB - BA)$. Prove that $(AB - BA)^n = O_n$.

24. Let $A, B \in M_n(\mathbb{R})$ such that $B^2 = I_n$ and $A^2 = AB + I_n$. Prove that

$$\det A \leq \left(\frac{1 + \sqrt{5}}{2} \right)^n.$$

25. Let $A \in M_n(\mathbb{C})$ be a matrix of rank 1. Prove that for any complex number λ ,

$$\det(\lambda I_n + A) = \lambda^n + \lambda^{n-1} \operatorname{tr} A.$$

26. Let $A, X \in M_n(\mathbb{R})$ be matrices such that $\operatorname{rank} X = 1$. Prove that

$$\det(A + X) \det(A - X) \leq (\det A)^2.$$

27. Let $n \geq 3$ be an odd integer, and let $A, B \in M_n(\mathbb{Z})$ be matrices such that $\det A \neq 0$ and $\operatorname{rank} B = 1$. Prove that the following statements are equivalent:

- (i) $\det A = \operatorname{tr} B \in \{-1, 1\}$;
- (ii) $\det(A^* + BA^{-1}) - \det(A^* - A^{-1}B) = 2$, where A^* is the adjugate matrix of A .

28. (Problem, Jeff Stuart, Pacific Lutheran University) Let $\{a_k\}_{k=1}^\infty$ be a sequence of complex numbers, and let b and c also be complex numbers. For every positive integer k , define the $k \times k$ matrix A_k by

$$A_k := \begin{pmatrix} a_1 & b & b & \cdots & b \\ c & a_2 & b & \cdots & b \\ c & c & a_3 & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c & c & c & \cdots & a_k \end{pmatrix}.$$

1. When all $a_j = a$ for some complex number a and when $b = c$, establish the following equation without using the formula in part 2 or the recursion in part 3: for $k \geq 1$, $\det(A_k) = (a - b)^{k-1}(a + kb - b)$.
2. When $k \geq 2$ and $b = c$, show that $\det(A_k) = 1 - b \frac{d}{db} \prod_{j=1}^k (a_j - b)$.
3. Show that for $k \geq 2$, $\det(A_{k+1}) = (a_k + a_{k+1} - b - c) \det(A_k) - (b - a_k)(c - a_k) \det(A_{k-1})$.

29. (Andrei Vila & Robert Rogojan, Romanian National Algebra Contest 2026, Problem 5)
Let $A, B \in M_n(\mathbb{C})$ and $P \in \mathbb{C}[X]$ such that $A^n = O_n$ and $P(0) = 0$. If $AB = BP(A)$, prove that $\det(A + B) = \det(B)$.

30. (Peter J. Costa, American Mathematical Monthly 101.5, May 1994, Problem 10387*)
Let $T_n = (t_{i,j})$ be the $n \times n$ matrix with $t_{i,j} = \tan((i + j - 1)x)$, i.e.,

$$T_n = \begin{pmatrix} \tan x & \tan 2x & \cdots & \tan nx \\ \tan 2x & \tan 3x & \cdots & \tan(n+1)x \\ \vdots & \vdots & \ddots & \vdots \\ \tan nx & \tan(n+1)x & \cdots & \tan(2n-1)x \end{pmatrix}.$$

Computer experiments suggest that $\det(T_n) = (-1)^{\lfloor n/2 \rfloor} \sec^n(nx)$. Prove this conjecture.

31. (Entries of a matrix inverse) Let n be a positive integer and let A be the $n \times n$ matrix defined by $A_{i,j} = 1$ if $\min(i, j)$ is odd, 2 if $\min(i, j)$ is even, i.e.,

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2 & 2 & \cdots & 2 \\ 1 & 2 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & 2 & \cdots & 2 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \end{pmatrix}.$$

Show that A^{-1} exists and determine the number of zero entries in A^{-1} .

32. (Yongge Tian, American Mathematical Monthly) Given two complex $n \times n$ positive-definite matrices A and B , let

$$C = \frac{1}{2}(A + B), \quad D = A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2};$$

the matrices C and D are the arithmetic mean and the geometric mean of A and B . Prove that

- (a) $\text{range}(C - D) = \text{range}(A - B)$.
(b) $\text{range} \begin{pmatrix} C & D \\ D & C \end{pmatrix} = \text{range} \begin{pmatrix} A & B \\ B & A \end{pmatrix}$.

33. (Tedszy Math, Problem 0095) Let $a_i \in \mathbb{R}$ and let $b_i \in \mathbb{R}$. Define the $n \times n$ matrix

$$M = \begin{pmatrix} a_1b_1 & a_1b_2 & a_1b_3 & \cdots & a_1b_n \\ a_2b_1 & a_2b_2 & a_2b_3 & \cdots & a_2b_n \\ a_3b_1 & a_3b_2 & a_3b_3 & \cdots & a_3b_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_nb_1 & a_nb_2 & a_nb_3 & \cdots & a_nb_n \end{pmatrix}.$$

What is the maximum rank of M ? When does it attain maximum rank?

34. (VMC 2026, Team Round, Problem 1) An $n \times n$ matrix A with real entries is called interesting if

$$A^2 = 2A^T - I \quad \text{and} \quad \text{tr}(A) + \text{tr}(A^2) = 0.$$

- a. Find all positive integers n such that an interesting matrix of size $n \times n$ exists.
- b. Find the possible determinants of A for each positive integer n for which interesting matrices of size $n \times n$ exist.

35. (Putnam 1999/B5) Let $n \geq 3$. Let A be the $n \times n$ matrix with $A_{jk} = \cos(2\pi(j+k)/n)$. Find $\det(I + A)$.

36. Let A and B be $n \times n$ matrices. Show that $\det(I + AB) = \det(I + BA)$.

37. Let A, B, C, D be $n \times n$ matrices such that $AC = CA$. Prove that

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - CB).$$

38. Let A be a square matrix with real entries. Show that $\det(A^2 + I) \geq 0$.

39. Let A, B, C be $n \times n$ real matrices that pairwise commute and $ABC = O$. Show that $\det(A^3 + B^3 + C^3) \det(A + B + C) \geq 0$.

40. Let A and B be two $n \times n$ real matrices that commute. Suppose that $\det(A + B) \geq 0$. Prove that $\det(A^k + B^k) \geq 0$ for all $k \geq 1$.

41. Let A be a real skew-symmetric square matrix (i.e., $A^T = -A$). Prove that $\det(I + tA^2) \geq 0$ for all real t .
42. Let A and B be real symmetric matrices with all eigenvalues strictly greater than 1. Let λ be a real eigenvalue of matrix AB . Prove that $|\lambda| > 1$.
43. Let V be a 10-dimensional real vector space and U_1 and U_2 two linear subspaces such that $U_1 \subseteq U_2$, $\dim_{\mathbb{R}} U_1 = 3$ and $\dim_{\mathbb{R}} U_2 = 6$. Let E be the set of all linear maps $T : V \rightarrow V$ which have U_1 and U_2 as invariant subspaces (i.e. $T(U_1) \subseteq U_1$ and $T(U_2) \subseteq U_2$). Calculate the dimension of E as a real vector space.
44. Determine all pairs (a, b) of real numbers for which there exists a unique symmetric 2×2 matrix M with real entries satisfying $\text{trace}(M) = a$ and $\det(M) = b$.
45. For any integer $n > 2$ and two $n \times n$ matrices with real entries A, B that satisfy the equation $A^{-1} + B^{-1} = (A + B)^{-1}$, prove that $\det(A) = \det(B)$. Does the same conclusion follow for matrices with complex entries?
46. (a) Show that for any $m \in \mathbb{N}$ there exists a real $m \times m$ matrix A such that $A^3 = A + I$ where I is the $m \times m$ identity matrix.
- (b) Show that $\det(A) > 0$ for every real $m \times m$ matrix satisfying $A^3 = A + I$.
47. Let A be the $n \times n$ matrix, whose (i, j) -th entry is $i + j$ for all $i, j = 1, 2, \dots, n$. What is the rank of A ?
48. Determine all complex numbers λ for which there exist a positive integer n and a real $n \times n$ matrix A such that $A^2 = A^T$ and λ is an eigenvalue of A .
49. Let A be a real $n \times n$ matrix such that $A^3 = 0$.
- (a) Prove that there is a unique real $n \times n$ matrix X that satisfies the equation $X + AX + XA^2 = A$.
- (b) Express X in terms of A .
50. (a) Let A be an $n \times n$, $n \geq 2$, symmetric invertible matrix with real positive elements. Show that $z_n \leq n^2 - 2n$, where z_n is the number of zero elements in A^{-1} .

(b) How many zero elements are there in the inverse of the $n \times n$ matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2 & \cdots & 2 \\ 1 & 2 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \end{pmatrix} ?$$

51. Let V be a real vector space, and let $f, f_1, f_2, \dots, f_k$ be linear maps from V to $\mathbb{R}$. Suppose that $f(x) = 0$ whenever $f_1(x) = f_2(x) = \cdots = f_k(x) = 0$. Prove that f is a linear combination of $f_1, f_2, \dots, f_k$.

52. Let A be a 3×3 real matrix such that the vectors Au and u are orthogonal for each column vector $u \in \mathbb{R}^3$. Prove that:

- (a) $A^T = -A$, where A^T denotes the transpose of A ;
- (b) there exists a vector $v \in \mathbb{R}^3$ such that $Au = v \times u$ for every $u \in \mathbb{R}^3$, where $v \times u$ denotes the vector product in $\mathbb{R}^3$.

53. Let $a_j = a_0 + jd$ for $j = 0, \dots, n$ where a_0, d are fixed real numbers. Put

$$A = \begin{pmatrix} a_0 & a_1 & a_2 & \cdots & a_n \\ a_1 & a_0 & a_1 & \cdots & a_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_0 \end{pmatrix}.$$

Calculate $\det(A)$.

54. Compute the determinant of the $n \times n$ matrix $A = [a_{ij}]$:

$$a_{ij} = \begin{cases} (-1)^{|i-j|}, & i \neq j \\ 2, & i = j. \end{cases}$$

55. Let A be a real 4×2 matrix and B be a real 2×4 matrix such that

$$AB = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}.$$

Find BA .

56. Let k be a positive integer. Find the smallest positive integer n for which there exist k nonzero vectors $v_1, \dots, v_k$ in $\mathbb{R}^n$ such that for every pair i, j of indices with $|i - j| > 1$ the vectors v_i and v_j are orthogonal.
57. Let A be a real $n \times n$ matrix and suppose that for every positive integer m there exists a real symmetric matrix B such that $2021B = A^m + B^2$. Prove that $|\det A| \leq 1$.
58. Let n be a positive integer. Find all $n \times n$ real matrices A with only real eigenvalues satisfying $A + A^k = A^T$ for some integer $k \geq n$.
59. Let $n \geq 2$ be an integer. What is the minimal and maximal possible rank of an $n \times n$ matrix whose n^2 entries are precisely the numbers $1, 2, \dots, n^2$?
60. Let M be an invertible matrix of dimension $2n \times 2n$, represented in block form as

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad M^{-1} = \begin{pmatrix} E & F \\ G & H \end{pmatrix}.$$

Show that $\det M \cdot \det H = \det A$.

61. Let A and B be $n \times n$ real matrices such that $\text{rank}(AB - BA + I) = 1$. Prove that

$$\text{trace}(ABAB) - \text{trace}(A^2B^2) = \frac{1}{2}n(n-1).$$

62. A four-digit number YEAR is called very good if the system

$$Yx + Ey + Az + Rw = Y$$

$$Rx + Yy + Ez + Aw = E$$

$$Ax + Ry + Yz + Ew = A$$

$$Ex + Ay + Rz + Yw = R$$

of linear equations in the variables x, y, z, w has at least two solutions. Find all very good YEARS in the 21st century.

63. Let $A = (a_{ij})_{i,j=1}^n$ be a symmetric $n \times n$ matrix with real entries, and let $\lambda_1, \lambda_2, \dots, \lambda_n$ denote its eigenvalues. Show that

$$\sum_{1 \leq i < j \leq n} a_{ii}a_{jj} \geq \sum_{1 \leq i < j \leq n} \lambda_i \lambda_j$$

and determine all matrices for which equality holds.

64. Let n be a fixed positive integer. Determine the smallest possible rank of an $n \times n$ matrix that has zeros along the main diagonal and strictly positive real numbers off the main diagonal.
65. Does there exist a real 3×3 matrix A such that $\text{tr}(A) = 0$ and $A^2 + A^T = I$?
66. Let $A_1, A_2, \dots, A_k$ be $n \times n$ idempotent complex matrices such that $A_i A_j = -A_j A_i$ for all $i \neq j$. Prove that at least one of the given matrices has rank $\leq n/k$.

67. Determine all rational numbers a for which the matrix

$$\begin{pmatrix} a & -a & -1 & 0 \\ a & -a & 0 & -1 \\ 1 & 0 & a & -a \\ 0 & 1 & a & -a \end{pmatrix}$$

is the square of a matrix with all rational entries.

68. Define the sequence $A_1, A_2, \dots$ of matrices by the following recurrence:

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_{n+1} = \begin{pmatrix} A_n & I_{2^n} \\ I_{2^n} & A_n \end{pmatrix} \quad (n = 1, 2, \dots)$$

where I_m is the $m \times m$ identity matrix. Prove that A_n has $n + 1$ distinct integer eigenvalues $\lambda_0 < \lambda_1 < \dots < \lambda_n$ with multiplicities $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$, respectively.

69. Let $A, B \in M_n(\mathbb{C})$ be two $n \times n$ matrices such that

$$A^2 B + B A^2 = 2 A B A.$$

Prove that there exists a positive integer k such that $(AB - BA)^k = 0$.

70. Let n be a positive integer, and consider the matrix $A = (a_{ij})_{1 \leq i, j \leq n}$, where

$$a_{ij} = \begin{cases} 1 & \text{if } i + j \text{ is a prime number} \\ 0 & \text{otherwise.} \end{cases}$$

Prove that $|\det A| = k^2$ for some integer k .

71. Let $n > 1$ be an odd positive integer and $A = (a_{ij})_{i, j=1 \dots n}$ be the $n \times n$ matrix with

$$a_{ij} = \begin{cases} 2 & \text{if } i = j \\ 1 & \text{if } i - j \equiv \pm 2 \pmod{n} \\ 0 & \text{otherwise.} \end{cases}$$

Find $\det(A)$.

72. Let A be an $n \times n$ -matrix with integer entries and $b_1, \dots, b_k$ be integers satisfying $\det A = b_1 \cdots b_k$. Prove that there exist $n \times n$ matrices $B_1, \dots, B_k$ with integer entries such that $A = B_1 \cdots B_k$ and $\det B_i = b_i$ for all $i = 1, \dots, k$.

73. Let v_0 be the zero vector in $\mathbb{R}^n$ and let $v_1, v_2, \dots, v_{n+1} \in \mathbb{R}^n$ be such that the Euclidean norm $|v_i - v_j|$ is rational for every $0 \leq i, j \leq n+1$. Prove that $v_1, \dots, v_{n+1}$ are linearly dependent over the rationals.

74. In the linear space of all real $n \times n$ matrices, find the maximum possible dimension of a linear subspace V such that $\forall X, Y \in V$, $\text{trace}(XY) = 0$.

75. For $n \geq 1$ let M be an $n \times n$ complex matrix with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ with multiplicities $m_1, m_2, \dots, m_k$ respectively. Consider the linear operator L_M defined by $L_M(X) = MX + XM^T$, for any complex $n \times n$ matrix X . Find its eigenvalues and their multiplicities.

76. Let A be an $n \times n$ real matrix such that $3A^3 = A^2 + A + I$. Show that the sequence A^k converges to an idempotent matrix.

77. Let A be an $n \times n$ complex matrix such that $A \neq \lambda I$ for all $\lambda \in \mathbb{C}$. Prove that A is similar to a matrix having at most one non-zero entry on the main diagonal.

78. Let A, B be square complex matrices of the same size such that $\text{rank}(AB - BA) = 1$. Show that $(AB - BA)^2 = 0$.

79. Let $\alpha \in \mathbb{R} \setminus \{0\}$ and suppose that F and G are linear maps (operators) from $\mathbb{R}^n$ into $\mathbb{R}^n$ satisfying $F \circ G - G \circ F = \alpha F$.

(a) Show that for all $k \in \mathbb{N}$ one has $F^k \circ G - G \circ F^k = \alpha k F^k$.

(b) Show that there exists $k \geq 1$ such that $F^k = 0$.

80. Let A be $n \times n$ diagonal matrix with characteristic polynomial

$$(x - c_1)^{d_1}(x - c_2)^{d_2} \cdots (x - c_k)^{d_k}$$

where $c_1, c_2, \dots, c_k$ are distinct. Let V be the space of all $n \times n$ matrices B such that $AB = BA$. Prove that the dimension of V is $d_1^2 + d_2^2 + \cdots + d_k^2$.

81. Let n be a positive integer. At most how many distinct unit vectors can be selected in $\mathbb{R}^n$ such that from any three of them, at least two are orthogonal?

82. Let A be a symmetric $m \times m$ matrix over the two-element field all of whose diagonal entries are zero. Prove that for every positive integer n each column of the matrix A^n has a zero entry.

83. Let A and B be real $n \times n$ matrices. Assume that there exist $n + 1$ different real numbers $t_1, t_2, \dots, t_{n+1}$ such that the matrices $C_i = A + t_i B$, $i = 1, 2, \dots, n + 1$ are nilpotent (i.e. $C_i^n = 0$). Show that both A and B are nilpotent.

84. The linear operator A on the vector space V is called an involution if $A^2 = E$ where E is the identity operator on V . Let $\dim V = n < \infty$.

(i) Prove that for every involution A on V there exists a basis of V consisting of eigenvectors of A .

(ii) Find the maximal number of distinct pairwise commuting involutions on V .

85. Determine all positive integers n for which there exist $n \times n$ real invertible matrices A and B that satisfy $AB - BA = B^2A$.

86. (a) Let $f : M_n \rightarrow \mathbb{R}$ be a linear map from the space $M_n = \mathbb{R}^{n^2}$ of real $n \times n$ matrices to the reals. Prove that there exists a unique matrix $C \in M_n$ such that $f(A) = \text{tr}(AC)$ for any $A \in M_n$.
- (b) Suppose in addition to (a) that $f(AB) = f(BA)$ for any $A, B \in M_n$. Prove that there exists $\lambda \in \mathbb{R}$ such that $f(A) = \lambda \cdot \text{tr}(A)$.
87. An $n \times n$ complex matrix A is called t-normal if $AA^T = A^T A$ where A^T is the transpose of A . For each n , determine the maximum dimension of a linear space of complex $n \times n$ matrices consisting of t-normal matrices.
88. Let A be an $n \times n$ complex matrix whose eigenvalues have absolute value at most 1. Prove that $\|A^n\| \leq n^{\ln 2} \|A\|^{n-1}$. (Here $\|B\| = \sup_{\|x\| \leq 1} \|Bx\|$ for every $n \times n$ matrix B and $\|x\| = \sqrt{\sum_{i=1}^n |x_i|^2}$ for every complex vector $x \in \mathbb{C}^n$.)
89. Let A be an $n \times n$ matrix with complex entries and suppose that $n > 1$. Prove that
- $$A\bar{A} = I_n \iff \exists S \in GL_n(\mathbb{C}) \text{ such that } A = S^{-1}\bar{S}.$$
90. Determine whether there exist an odd positive integer n and $n \times n$ matrices A and B with integer entries, that satisfy the following conditions:
- (a) $\det(B) = 1$
- (b) $AB = BA$
- (c) $A^4 + 4A^2B^2 + 16B^4 = 2019I$
91. For each positive integer k , find the smallest number n_k for which there exist real $n_k \times n_k$ matrices $A_1, A_2, \dots, A_k$ such that all of the following conditions hold:
- (1) $A_1^2 = A_2^2 = \dots = A_k^2 = 0$,
- (2) $A_i A_j = A_j A_i$ for all $1 \leq i, j \leq k$, and
- (3) $A_1 A_2 \dots A_k \neq 0$.
92. For an $m \times m$ real matrix A , e^A is defined as $\sum_{n=0}^{\infty} \frac{1}{n!} A^n$. (The sum is convergent for all matrices.) Prove or disprove that for all real polynomials p and $m \times m$ real matrices A and B , $p(e^A B)$ is nilpotent if and only if $p(e^B A)$ is nilpotent.

93. For $n \geq 0$ define matrices A_n and B_n as follows: $A_0 = B_0 = (1)$ and for every $n > 0$,

$$A_n = \begin{pmatrix} A_{n-1} & A_{n-1} \\ A_{n-1} & B_{n-1} \end{pmatrix}, \quad B_n = \begin{pmatrix} A_{n-1} & A_{n-1} \\ A_{n-1} & 0 \end{pmatrix}.$$

Denote the sum of all elements of a matrix M by $S(M)$. Prove that $S(A_n^{k-1}) = S(A_k^{n-1})$ for every $n, k \geq 1$.

94. Let A_i, B_i, S_i ($i = 1, 2, 3$) be invertible real 2×2 matrices such that:

(1) not all A_i have a common real eigenvector;

(2) $A_i = S_i^{-1} B_i S_i$ for all $i = 1, 2, 3$;

(3) $A_1 A_2 A_3 = B_1 B_2 B_3 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Prove that there is an invertible real 2×2 matrix S such that $A_i = S^{-1} B_i S$ for all $i = 1, 2, 3$.

95. Let $A^{n \times n}$ and $B^{n \times n}$ be two matrices which differ only by one row/column. Show that $\det(A + B) = 2^{n-1} \{\det(A) + \det(B)\}$.

96. Let $A = (a_{ij})$ and $B = (b_{ij})$ be two square matrices of the same dimension.

(a) Let $a_{ij} = \rho^{|i-j|}$; show that $\det(A) = (1 - \rho^2)^{n-1}$.

(b) Suppose now $b_{ij} = \rho^{i-j} a_{ij}$. Find $\det(B)$ in terms of ρ .

97. Consider the matrix

$$B^{n \times n}(x) = \begin{pmatrix} 1 & x & 0 & \cdots & 0 \\ x & 1 & x & \cdots & 0 \\ 0 & x & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}, \quad x \in \mathbb{R};$$

find $\det(B^{n \times n}(x))$ for any $x \in \mathbb{R}$. What if $x = \frac{1}{2}$?

98. Prove that

$$\begin{vmatrix} 1+a_1 & a_2 & \cdots & a_n \\ a_1 & 1+a_2 & \cdots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \cdots & 1+a_n \end{vmatrix} = 1 + \sum_{i=1}^n a_i.$$

99. Let $g(x) = \prod_{i=1}^n (u_i - x)$ and let

$$\Delta_n = \begin{vmatrix} u_1 & a & a & \cdots & a \\ b & u_2 & a & \cdots & a \\ b & b & u_3 & \cdots & a \\ \vdots & \vdots & \vdots & \ddots & a \\ b & b & b & \cdots & u_n \end{vmatrix}.$$

(a) Show that if $a \neq b$ then $\Delta_n = \frac{bg(a) - ag(b)}{b - a}$.

(b) Also show that if $a = b$ then $\Delta_n = a \sum_{i=1}^n g_i(a) + u_n g_n(a)$ where $g_i(a) = \frac{g(a)}{u_i - a}$ for $i = 1, \dots, n$.

(c) Use (b) to evaluate $\det((a - b)I_n + b \mathbf{1}_n \mathbf{1}_n^T)$.

100. Show that if $a \neq b$ then

$$\begin{vmatrix} a+b & ab & 0 & 0 & \cdots & 0 \\ 1 & a+b & ab & 0 & \cdots & 0 \\ 0 & 1 & a+b & ab & \cdots & 0 \\ 0 & 0 & 1 & a+b & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & a+b \end{vmatrix} = \frac{a^{n+1} - b^{n+1}}{a - b};$$

what if $a = b$?

101. Let $p < n$ be any two positive integers. Prove that $\det(\Delta_{n,p}) = 1$ where $\Delta_{n,p}$ is the matrix with (i, j) -th entry $\delta_{ij}^{n,p} = \binom{n+i-1}{j-1}$ for $1 \leq i, j \leq p+1$.

102. Prove that the determinant of a circulant matrix is of the following form:

$$\begin{vmatrix} c_1 & c_2 & c_3 & \cdots & c_n \\ c_n & c_1 & c_2 & \cdots & c_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_2 & c_3 & c_4 & \cdots & c_1 \end{vmatrix} = (-1)^{n-1} \prod_{k=1}^n \left(\sum_{j=1}^n \zeta^{k(j-1)} c_j \right)$$

where $\zeta = e^{2\pi i/n}$.

103. Given distinct integers $x_1, x_2, \dots, x_n$, prove that $\prod_{i>j} (x_i - x_j)$ is divisible by $\prod_{k=1}^{n-1} k!$.

104. Let $f_{ij}(t)$ be a differentiable function of t . Then show that

$$\frac{d}{dt} \begin{vmatrix} f_{11}(t) & \cdots & f_{1n}(t) \\ \vdots & \ddots & \vdots \\ f_{n1}(t) & \cdots & f_{nn}(t) \end{vmatrix} = \sum_{j=1}^n \det(F_j(t))$$

where $F_j(t)$ is the matrix with the j -th row differentiated.

105. Compute the determinant of the $n \times n$ matrix $B = (b_{ij})$ where

$$b_{ij} = \begin{cases} (-1)^{|i-j|} & i \neq j \\ 2 & i = j. \end{cases}$$

106. If A is an $n \times n$ matrix over $\mathbb{C}$ then show that $|\det(A)| \leq \prod_{i=1}^n \left(\sum_{j=1}^n |a_{ij}| \right)$.

107. Show that the determinant of the Cauchy matrix $A = (a_{ij})$ where $a_{ij} = \frac{1}{x_i + y_j}$ is of the following form:

$$\det(A) = \frac{\prod_{i>j} (x_i - x_j)(y_i - y_j)}{\prod_{i,j} (x_i + y_j)}.$$

108. (The Binet–Cauchy Formula) Let A and B be matrices of size $n \times m$ and $m \times n$ where $n \leq m$. Then show that

$$\det(AB) = \sum_{1 \leq k_1 < k_2 < \cdots < k_n \leq m} A^{k_1, k_2, \dots, k_n} B_{k_1, k_2, \dots, k_n}$$

where $A^{k_1, \dots, k_n}$ is the minor obtained from A using columns $k_1, \dots, k_n$ and $B_{k_1, \dots, k_n}$ is the minor obtained from B using rows $k_1, \dots, k_n$.

109. Let X and Y be $n \times n$ matrices. Prove that $\det(I_n - XY) = \det(I_n - YX)$.

110. Let M be an $n \times n$ matrix such that $M + M^T = O_n$. Prove that for all $\epsilon \in \mathbb{R}$, $\det(I_n + \epsilon M^2) \geq 0$.

111. Let $S_i = \sum_{j=1}^n b_{ij}$. Prove that

$$\begin{vmatrix} S_1 - b_{11} & \cdots & S_1 - b_{1n} \\ \vdots & \ddots & \vdots \\ S_n - b_{n1} & \cdots & S_n - b_{nn} \end{vmatrix} = (-1)^{n-1}(n-1) \det(B)$$

where $B = (b_{ij})_{i,j=1}^n$.

112. Prove that

$$\begin{vmatrix} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & b_n \\ c_1 & \cdots & c_n & 0 \end{vmatrix} = - \sum_{i,j} b_i c_j A_{ij}$$

where A_{ij} is the cofactor of a_{ij} in A .

113. Let u and v be columns of length n . Prove that $\text{adj}(I - uv^T) = uv^T + (1 - v^T u)I$.

114. Let A, B, C, D be $n \times n$ matrices such that $ABCD = I$. Show that $ABCD = DABC = CDAB = BCDA = I$.

115. Find the inverse of the matrix $A = (a_{ij})_{i,j=1}^n$ where $a_{ij} = \min(i, j)$ for $i, j = 1, \dots, n$.

116. Let $A, B, C, D \in M_{n \times n}(\mathbb{R})$ and the matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ has rank n . Show that

$$\begin{vmatrix} \det(A) & \det(B) \\ \det(C) & \det(D) \end{vmatrix} = 0;$$

also if A is invertible, then $D = CA^{-1}B$.

117. Let A and B be $n \times n$ complex matrices. Show that

$$\begin{vmatrix} A & B \\ B & A \end{vmatrix} = \det(A+B) \det(A-B).$$

Let $E = A + B$ and $F = A - B$. If E and F are invertible then

$$\begin{pmatrix} A & B \\ B & A \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} E^{-1} + F^{-1} & E^{-1} - F^{-1} \\ E^{-1} - F^{-1} & E^{-1} + F^{-1} \end{pmatrix}.$$

118. Let $A \in M_n(\mathbb{R})$ be a skew-symmetric matrix. Show that $\det(A) \geq 0$.

119. Let $A \in M_n(\mathbb{R})$ such that $A^3 = A + I_n$. Show that $\det(A) > 0$.

120. Show that if $A, B \in M_n(\mathbb{R})$ such that $AB = BA$ then $\det(A^2 + B^2) \geq 0$.

121. Let $A, B \in M_2(\mathbb{R})$ such that $AB = BA$ and $\det(A^2 + B^2) = 0$. Prove that $\det(A) = \det(B)$.

122. Let $A \in M_2(\mathbb{R})$ such that $\det(A) = -1$. Show that $\det(A^2 + I_2) \geq 4$. When does the equality hold?

123. Let $A, B \in M_3(\mathbb{C})$ such that $\det(A) = \det(B) = 1$. Show that $\det(A + \sqrt{2}B) \neq 0$.

124. Let $A, B \in M_2(\mathbb{R})$. Show that $\det((AB + BA)^4 + (AB - BA)^4) \geq 0$.

125. (a) Let A be a matrix from $M_2(\mathbb{C})$, $A \neq aI_2$ for any $a \in \mathbb{C}$. Prove that the matrix X from $M_2(\mathbb{C})$ commutes with A , that is, $AX = XA$, if and only if there exist two complex numbers α, α' such that $X = \alpha A + \alpha' I_2$.

(b) Let A, B and C be matrices from $M_2(\mathbb{C})$, such that $AB \neq BA$, $AC = CA$ and $BC = CB$. Prove that C commutes with all matrices from $M_2(\mathbb{C})$.

126. Let $A, B, C \in M_3(\mathbb{R})$ such that $\det(A) = \det(B) = \det(C)$ and $\det(A + iB) = \det(C + iA)$. Prove that $\det(A + B) = \det(C + A)$.

127. Let $A \in M_n(\mathbb{R})$. If $A^T A = I_n$, prove that: a) $|\text{Tr}(A)| \leq n$; b) If n is odd, then $\det(A^2 - I_n) = 0$.
128. Consider the matrices $A, B \in M_3(\mathbb{C})$ with $A = -A^T$ and $B = B^T$. Prove that if the polynomial function defined by $f(x) = \det(A + xB)$ has a multiple root, then $\det(A + B) = \det B$.
129. Let $A, B \in M_2(\mathbb{C})$ be two non-zero matrices such that $AB + BA = O_2$ and $\det(A + B) = 0$. Prove that A and B have null traces.
130. Let A, B be 2×2 matrices with real elements such that $AB = A^2 B^2 - (AB)^2$ and $\det(B) = 2$.
- (a) Prove that the matrix A is not invertible.
- (b) Calculate $\det(A + 2B) - \det(B + 2A)$.
131. (a) Give an example of matrices A and B from $M_2(\mathbb{R})$ such that $A^2 + B^2 = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$.
- (b) Let A and B be matrices from $M_2(\mathbb{R})$ such that $A^2 + B^2 = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$. Show that $AB \neq BA$.
132. Show that the function $f : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$, $f(X) = X^n$, is not injective nor surjective for all $n \geq 2$.
133. Let $A, B \in M_2(\mathbb{R})$. Prove that $\det(A^2 + B^2) \geq \det(AB - BA)$.
134. Let $A, B \in M_2(\mathbb{R})$ such that $\det(AB + BA) \leq 0$. Show that $\det(A^2 + B^2) \geq 0$.
135. Let $A \in M_3(\mathbb{R})$ such that $\text{r}(A) = \text{tr}(A^2) = 0$. Show that $\det(A^2 + I_3) = (\det(A))^2 + 1$.
136. Let $A \in M_3(\mathbb{C})$ such that $\text{tr}(A) = \text{tr}(A^2) = 0$. Show that if $|\det(A)| < 1$ then the matrix $I_3 + A^n$ is invertible for all $n \geq 1$.
137. Let $A, B \in M_2(\mathbb{R})$ be two matrices with positive entries. Show that $(AB)^2 = (BA)^2$ if and only if $AB = BA$.

138. Let $A \in M_2(\mathbb{R})$ such that $\det(A) \geq 0$. Show that

$$\det(A^n + I_2) + \det(A^n - I_2) \geq \frac{1}{2^{n-2}}(\det(A) + 1)^n.$$

139. Let $A \in M_2(\mathbb{R})$ be a matrix such that $A \neq \operatorname{adj}(A)$ and $A^3 = (\operatorname{adj}(A))^3$. Show that $\det(A) = \operatorname{tr}^2(A)$.

140. Find all polynomials P with real coefficients with the following property: for any two matrices $A, B \in M_2(\mathbb{R})$ with $A \neq B$ we have $P(A) \neq P(B)$.

141. Let $A, B \in M_2(\mathbb{Z})$ with $AB = BA$ such that $\det(A) = \det(B) = 0$. Show that $\det(A^3 + B^3)$ is the cube of an integer.

142. Let $n \geq 3$ be a positive integer. Determine $X \in M_2(\mathbb{R})$ such that

$$X^n + X^{n-2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

143. Let $A \in M_2(\mathbb{R})$ such that $\operatorname{tr}(A) > 2$. Show that $A^n \neq I_2$ for all $n \geq 1$.

144. (a) Let $A \in M_3(\mathbb{C})$ such that $A^2 = O_3$. Show that $\operatorname{tr}(A) = 0$.

(b) Let $A \in M_3(\mathbb{C})$ such that $\operatorname{tr}(A) = \operatorname{tr}(A^2) = 0$. Show that $\det(A) = 0$.

145. Show that the equation $X^n = I_2$ has at least n solutions in $M_2(\mathbb{R})$ with $n \geq 1$.

146. Find all matrices $A \in M_2(\mathbb{Z})$ such that $\det(A^3 + I_2) = 1$.

147. Let $A, B \in M_2(\mathbb{Z})$ such that $AB = BA$ and $\det(B) = 1$. Show that if $\det(A^3 + B^3) = 1$, then $A^2 = O_2$.

148. Let $A, B \in M_2(\mathbb{R})$ with $AB = BA$. If there exists a positive integer n such that $\det(A^{2n} + B^{2n}) = 0$ then the matrices A and B are simultaneously invertible or non-invertible.

149. Let $A, B \in M_2(\mathbb{R})$ such that $A^2 + B^2 = 2AB$. Prove that $AB = BA$ and $\operatorname{tr}(A) = \operatorname{tr}(B)$.

150. Show that if $A, B, C \in M_3(\mathbb{C})$ verify $A^3 = B^3 = C^3 = O_3$ and they commute pairwise, then $ABC = O_3$.
151. Let $A, B \in M_3(\mathbb{C})$. Show that $\det(AB - BA) = \operatorname{tr}(AB(BA - AB)BA)$.
152. Let $A, B \in M_n(\mathbb{R})$, $A \neq B$, such that $A^3 = B^3$ and $A^2B = AB^2$. Is it possible that the matrix $A^2 + B^2$ is invertible?
153. Let $A, B \in M_n(\mathbb{R})$ such that $(AB)^2 = O_n$. Is it true that $(BA)^2 = O_n$?
154. Let $A, B \in M_n(\mathbb{R})$ such that $AA^T = I_n$ and $BB^T = I_n$, with n odd. Show that at least one of the matrices $A + B$ and $A - B$ is singular.
155. Let $A, B \in M_n(\mathbb{R})$ with $B^2 = O_n$. Show that $\det(AB + BA + I_n) \geq 0$.
156. (a) Find two matrices $A, B \in M_2(\mathbb{C})$ such that $A^2 + B^2 = I_2$ and $AB - BA$ is nonsingular.
 (b) Let n be odd and $A, B \in M_n(\mathbb{C})$ such that $A^2 + B^2 = I_n$. Show that $\det(AB - BA) \geq 0$.
157. Let $A, B, C \in M_2(\mathbb{R})$ be matrices that commute pairwise such that $\det(A) = 0$. Show that $\det(A^2 + B^2 + C^2) \geq 0$.
158. Let $A \in M_2(\mathbb{Z})$ such that there exists $n \in \mathbb{N}$, $n \geq 1$, with $\gcd(n, 6) = 1$ and $A^n = I_2$. Show that $A = I_2$.
159. Show that if $A \in M_3(\mathbb{Q})$ such that $A^8 = I_3$, then $A^4 = I_3$.
160. Let $A \in M_n(\mathbb{R})$ such that $|\operatorname{tr}(A)| > n$. Show that $A^k \neq A^p$ for all $k, p \in \mathbb{N}$, $k \neq p$.
161. Let A be an $n \times n$ matrix with real entries. Show that $\operatorname{tr}(A^k) = 0$ for all $k = 0, 1, \dots, n$ if and only if $A^n = O_n$.
162. (a) Are there matrices $A, B \in M_3(\mathbb{C})$ such that $(AB - BA)^2 = I_3$?
 (b) Are there matrices $A, B \in M_3(\mathbb{C})$ such that $(AB - BA)^{1993} = I_3$?

163. Let A and B be two matrices from $M_3(\mathbb{C})$ such that $(AB)^2 = A^2B^2$ and $(BA)^2 = B^2A^2$. Prove that $(AB - BA)^3 = O_3$.
164. Let A, B be two matrices from $M_3(\mathbb{C})$ such that $A^2 = AB + BA$. Prove that the matrix $AB - BA$ is singular.
165. Let $A, B \in M_k(\mathbb{R})$ with $AB = BA$. Show that $\det(A + B) \geq 0$ if and only if $\det(A^n + B^n) \geq 0$ for all $n \geq 1$.
166. Let r, s be odd prime numbers and $A, B \in M_n(\mathbb{C})$ such that $AB = BA$, $A^r = I_n$ and $B^s = I_n$. Show that the matrix $A + B$ is invertible.
167. Let $n, k \geq 1$ and $A_1, A_2, \dots, A_k \in M_n(\mathbb{R})$. Show that $\det\left(\sum_{i=1}^k A_i^T A_i\right) \geq 0$.
168. Let $A, B, C, D \in M_n(\mathbb{C})$ where A and C are invertible. If $A^k B = C^k D$ for all $k \geq 1$ then show that $B = D$.
169. Let $A_0, A_1, \dots, A_n \in M_2(\mathbb{R})$, $n \geq 2$, be $n + 1$ matrices such that $A_0 \neq a \cdot A_2$ for any real a and $A_0 A_k = A_k A_0$ for all $k = 1, 2, \dots, n$. Show that
- (a) $\det\left(\sum_{k=1}^n A_k^2\right) \geq 0$;
- (b) If $\det\left(\sum_{k=1}^n A_k^2\right) = 0$ and $A_2 \neq a \cdot A_1$ for any real number a , then $\sum_{k=1}^n A_k^2 = O_n$.
170. Let $A \in M_2(\mathbb{C})$ and $C(A) = \{B \in M_2(\mathbb{C}) : AB = BA\}$. Show that $|\det(A + B)| \geq |\det(B)|$ for any $B \in C(A)$ if and only if $A^2 = O_2$.
171. Let $M = \{A \in M_2(\mathbb{C}) : \det(A - zI_2) = 0 \Rightarrow |z| < 1\}$. Show that if $A, B \in M$ and $AB = BA$, then $AB \in M$.
172. Let $A \in M_n(\mathbb{Z})$. Show that $\det(A^4 + I_n) \neq 13$.
173. Let $A, B \in M_n(\mathbb{C})$. Show that if the equation $AX + XB = C$, with $X, C \in M_n(\mathbb{C})$, has a solution for all $C \in M_n(\mathbb{C})$, then $\det(A) \neq 0$ or $\det(B) \neq 0$.

174. Let $A \in M_n(\mathbb{Z})$, $m > 1$, such that $\gcd(m, \det(A)) = 1$. Show that the matrix $A^2 + A + m \cdot I_n$ is nonsingular.
175. Let $A \in M_2(\mathbb{Z})$ different from the null matrix. Show that the following assertions are equivalent: (i) $4 \det(A) = (\operatorname{tr}(A))^2$ and $A \neq k \cdot I_2$ for $k \in \mathbb{Z}$; (ii) for any matrix $B \in M_2(\mathbb{Z})$ which commutes with A , $\det(A^2 + B^2)$ is the square of an integer.
176. Let $f : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$, $n \geq 2$, be a function such that $f(X \cdot Y) = f(X) \cdot f(Y)$, for all $X, Y \in M_n(\mathbb{C})$. Show that f is not a bijection.
177. Consider the matrices $A, B \in M_n(\mathbb{C})$ and the polynomial $f \in \mathbb{C}[X]$, $f(X) = \det(A + X \cdot B)$. Show that the degree of the polynomial f is less than or equal to the rank of the matrix B .
178. (a) Let A be a square matrix of size 3, with real entries. Show that if f is a polynomial with real coefficients, but with no real roots, then $f(A) \neq O_3$.
 (b) Show that there exists a positive integer n such that $(A - \operatorname{adj}(A))^{2n} + A^{2n} + (\operatorname{adj}(A))^{2n} = O_3$ if and only if $\det(A) = 0$.
179. (a) Show that any matrix $A \in M_4(\mathbb{C})$ can be written as the sum of four matrices $B_i \in M_4(\mathbb{C})$, $i = 1, 2, 3, 4$, of rank 1.
 (b) Show that I_4 cannot be written as the sum of four matrices of rank 1.
180. Let $A, B \in M_n(\mathbb{C})$ such that there exists $k > 1$ with $B^k = O_n$. Show that if $AB = BA$, then $\det(A + B) = \det(A)$.
181. Let A, B be two $n \times n$ matrices with complex entries, and p, q fixed positive integers such that $AB = BA$ and $A^p = B^q = O_n$. Show that the matrix $A + B + I_n$ is invertible.
182. Let $m, n > 2$ be integers. The matrices $A_1, A_2, \dots, A_m \in M_n(\mathbb{R})$ are not all nilpotent. Prove that there is an integer $k > 0$ such that $A_1^k + A_2^k + \dots + A_m^k \neq O_n$.
183. Let A, B be 3×3 matrices with real entries such that $AB = O_3$.
 (a) Prove that $f : \mathbb{C} \rightarrow \mathbb{C}$, $f(x) = \det(A^2 + B^2 + x \cdot BA)$ is a polynomial function of degree not greater than 2.

- (b) Prove that $\det(A^2 + B^2) \geq 0$.
184. Consider a positive integer n and A, B two matrices in $M_n(\mathbb{C})$ such that $A^2 + B^2 = 2AB$. Prove that:
- (a) The additive commutator $AB - BA$ is not invertible;
 - (b) If the matrix $A - B$ has rank 1, then matrices A and B commute.
185. Let A be an invertible matrix in $M_4(\mathbb{R})$ such that $\text{tr}(A) = \text{tr}(\text{adj}(A)) \neq 0$. Prove that the matrix $A^2 + I_4$ is singular if and only if there exists a nonzero matrix B in $M_4(\mathbb{R})$, so that $AB = -BA$.
186. Consider two nonnegative integers n and k such that $n \geq 2$ and $1 \leq k \leq n-1$. Suppose that a matrix $A \in M_n(\mathbb{C})$ has exactly k minors of order $n-1$ which are null. Prove that $\det(A) \neq 0$.
187. Let A be a non-invertible square matrix of size n , $n > 2$, with real entries, and let $\text{adj}(A)$ be the adjugate of A . Prove that $\text{tr}(\text{adj}(A)) \neq -1$ if and only if the matrix $I_n + \text{adj}(A)$ is invertible.
188. Consider $1 \leq a_1 < a_2 < a_3 < a_4$ and $x_1 < x_2 < x_3 < x_4$ be real numbers and $M = (a_i^{x_j})_{1 \leq i, j \leq 4}$ be a square matrix of size 4. Show that $\det(M) > 0$.
189. Let $S(X)$ be the sum of all the entries of the matrix $X \in M_n(\mathbb{R})$. Show that for any matrices $A_1, A_2, \dots, A_k \in M_n(\mathbb{R})$ we have the following inequality:
- $$S^2(A_1 A_2^T) \cdot S^2(A_2 A_3^T) \cdots S^2(A_k A_1^T) \leq S^2(A_1 A_1^T) \cdot S^2(A_2 A_2^T) \cdots S^2(A_k A_k^T).$$
190. Let $n > k$ and let $A_1, A_2, \dots, A_k$ be real $n \times n$ matrices of rank $n-1$. Prove that $A_1 A_2 \cdots A_k \neq 0$ (has rank strictly less than n ; state and prove the appropriate rank bound).
191. Two matrices A, B of the same size having real entries satisfy $A^{2002} = B^{2003} = I_n$ and $AB = BA$. Show that $A + B + I_n$ is invertible.

192. Let A be an $n \times n$ matrix of real numbers for some $n \geq 1$. For each positive integer k , let $A^{[k]}$ be the matrix obtained by raising each entry to the k -th power. Show that if $A^k = A^{[k]}$ for $k = 1, 2, \dots, n+1$, then $A^k = A^{[k]}$ for all $k \geq 1$.
193. Let A and B be 2×2 matrices with integer entries such that $A, A+B, A+2B, A+3B$, and $A+4B$ are all invertible matrices whose inverses have integer entries. Show that $A+5B$ is invertible and that its inverse has integer entries.
194. Let $A, B, C \in M_n(\mathbb{R})$ such that $ABC = O_n$ and $\text{rank}(B) = 1$. Prove that $AB = O_n$ or $BC = O_n$.
195. Let $A, B \in M_n(\mathbb{C})$ such that $AB = BA$ and $\det(B) \neq 0$.
- If $|\det(A + zB)| = 1$ for any $z \in \mathbb{C}$ such that $|z| = 1$ then $A^n = O_n$.
 - Is the statement from (a) still true if $AB \neq BA$?
196. Let A be an $n \times n$ matrix with complex elements. Prove that $A^{-1} = A$ if and only if there exists an invertible matrix B with complex elements such that $A = B^{-1}\overline{B}$.
197. Let $A = (a_{ij})_{1 \leq i, j \leq n}$ be a real $n \times n$ matrix, such that $a_{ij} + a_{ji} = 0$ for all i, j . Prove that for all non-negative real numbers x, y we have
- $$\det(A + xI_n) \cdot \det(A + yI_n) \geq \det(A + \sqrt{xy}I_n)^2.$$
198. Let $A, B \in M_2(\mathbb{R})$ that satisfy $A^2 + B^2 = AB$. Prove that $(AB - BA)^2 = O_2$.
199. Let $n \in \mathbb{N}$, $n \geq 2$.
- Give an example of two matrices $A, B \in M_n(\mathbb{C})$ such that $\text{rank}(AB) - \text{rank}(BA) = \lfloor n/2 \rfloor$.
 - Prove that for all matrices $X, Y \in M_n(\mathbb{C})$ we have $\text{rank}(XY) - \text{rank}(YX) \leq \lfloor n/2 \rfloor$.
200. (a) Find a matrix $A \in M_3(\mathbb{C})$ such that $A^2 \neq O_3$ and $A^3 = O_3$.
- (b) Let $n, p \in \{2, 3\}$. Prove that if there is a bijective function $f : M_n(\mathbb{C}) \rightarrow M_p(\mathbb{C})$ such that $f(XY) = f(X)f(Y)$ for all $X, Y \in M_n(\mathbb{C})$, then $n = p$.

201. (a) Let $A, B \in M_3(\mathbb{R})$ such that $\text{rank}(A) > \text{rank}(B)$. Prove that $\text{rank}(A^2) \geq \text{rank}(B^2)$.
- (b) Find the non-constant polynomials $f \in \mathbb{R}[X]$ such that for all $A, B \in M_4(\mathbb{R})$ with $\text{rank}(A) > \text{rank}(B)$, we have $\text{rank}(f(A)) > \text{rank}(f(B))$.

202. Let $p \geq 2$ be an integer and A an $n \times n$ matrix with real elements such that $A^{p+1} = A$.

- (a) Prove that $\text{rank}(A) + \text{rank}(I_n - A^p) = n$.
- (b) Prove that if p is prime then $\text{rank}(I_n - A) = \text{rank}(I_n - A^2) = \cdots = \text{rank}(I_n - A^{p-1})$.

203. Let $A, B \in M_n(\mathbb{R})$. Prove that $\text{rank}(A) + \text{rank}(B) < n$ if and only if there exists an invertible matrix $X \in M_n(\mathbb{R})$ such that $AXB = O_n$.

204. Let $S_1, S_2, \dots, S_{2^n-1}$ be the nonempty subsets of $\{1, 2, \dots, n\}$ in some order, and let M be the $(2^n - 1) \times (2^n - 1)$ matrix whose (i, j) entry is

$$m_{ij} = \begin{cases} 0 & \text{if } S_i \cap S_j = \emptyset \\ 1 & \text{otherwise.} \end{cases}$$

Determine $\det(M)$.

205. Let P be the set of vectors defined by $P = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \mid 0 \leq a \leq 2, 0 \leq b \leq 100, a, b \in \mathbb{Z} \right\}$. Find all $v \in P$ such that $P \setminus \{v\}$ can be partitioned into two sets of equal size and equal sum.

206. Let n be a positive integer. Suppose that A, B, M are $n \times n$ matrices with real entries such that $AM = MB$ and such that A and B have the same characteristic polynomial. Prove that $\det(A - MX) = \det(B - XM)$ for every $n \times n$ matrix X with real entries.

207. Let S be the set of all 2×2 matrices $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ whose entries a, b, c, d (in that order) form an arithmetic progression. Find all matrices M in S for which there is some integer $k > 1$ such that M^k is also in S .

208. Let A be the $n \times n$ matrix whose entry in the i -th row and j -th column is $\frac{1}{\min(i, j)}$ for $1 \leq i, j \leq n$. Compute $\det(A)$.

209. Let n be a positive integer. What is the largest k for which there exist $n \times n$ matrices $M_1, \dots, M_k$ and $N_1, \dots, N_k$ with real entries such that, for all i and j , the matrix product $M_i N_j$ has a zero entry somewhere on its main diagonal if and only if $i \neq j$.
210. Let A be an $m \times n$ matrix with rational entries. Suppose that there are at least $m + n$ distinct prime numbers among the absolute values of the entries of A . Show that the rank of A is at least 2.
211. For which positive integers n is there an $n \times n$ matrix with integer entries such that every dot product of a row with itself is even, while every dot product of two different rows is odd?
212. Let d_n be the determinant of the $n \times n$ matrix whose entries, from left to right and then from top to bottom, are $\cos 1, \cos 2, \dots, \cos n^2$. (For example,

$$d_3 = \begin{vmatrix} \cos 1 & \cos 2 & \cos 3 \\ \cos 4 & \cos 5 & \cos 6 \\ \cos 7 & \cos 8 & \cos 9 \end{vmatrix},$$

where the argument of $\cos$ is always in radians, not degrees.) Evaluate $\lim_{n \rightarrow \infty} d_n$.

213. Say that a polynomial with real coefficients in two variables x, y is balanced if the average value of the polynomial on each circle centered at the origin is 0. The balanced polynomials of degree at most 2009 form a vector space V over $\mathbb{R}$. Find the dimension of V .
214. Let Z denote the set of points in $\mathbb{R}^n$ whose coordinates are 0 or 1. (Thus Z has 2^n elements, which are vertices of a hypercube in $\mathbb{R}^n$.) Given a vector subspace V of $\mathbb{R}^n$, let $Z(V)$ denote the number of members of Z that lie in V . Let k be given, $0 < k < n$. Find the maximum, over all vector subspaces $V \subset \mathbb{R}^n$ of dimension k , of the number of points in $V \cap Z$.
215. Let H be an $n \times n$ matrix all of whose entries are ± 1 and whose rows are mutually orthogonal. Suppose H has an $a \times b$ submatrix whose entries are all 1. Show that $ab \leq n$.

216. Define a sequence $\{u_n\}_{n=0}^\infty$ by $u_0 = u_1 = u_2 = 1$, and thereafter by the condition that

$$\det \begin{pmatrix} u_n & u_{n+1} \\ u_{n+2} & u_{n+3} \end{pmatrix} = n!$$

for all $n \geq 0$. Show that u_n is an integer for all n . (By convention, $0! = 1$.)

217. For an integer $n \geq 3$ let $\theta = 2\pi/n$. Evaluate the determinant of the $n \times n$ matrix $I + A$, where I is the $n \times n$ identity matrix and $A = (a_{jk})$ has entries $a_{jk} = \cos(j\theta + k\theta)$ for all j, k .

218. For any square matrix A , we can define $\sin A$ by the usual power series $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} A^{2n+1}$. Prove or disprove: there exists a 2×2 matrix A with real entries such that

$$\sin A = \begin{pmatrix} 1 & 1996 \\ 0 & 1 \end{pmatrix}.$$

219. To each positive integer with n^2 decimal digits we associate the determinant of the matrix obtained by writing the digits in order across the rows. For example, for $n = 2$, to the integer 8617 we associate $\det \begin{pmatrix} 8 & 6 \\ 1 & 7 \end{pmatrix} = 50$. Find, as a function of n , the sum of all the determinants associated with n^2 -digit integers. (Leading digits are assumed to be nonzero; for example, for $n = 2$ there are 9000 determinants.)

220. For $n > 1$ let d_n be the greatest common divisor of the entries of $A^n - I$, where $A = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Show that $\lim_{n \rightarrow \infty} d_n = \infty$.

221. Let D_n denote the value of the $(n-1) \times (n-1)$ determinant

$$D_n = \begin{vmatrix} 1 & 1 & 1 & \cdots \\ 1 & 1 & 1 & \cdots \\ 1 & 1 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix}.$$

Is the set $\{D_n/n! : n \geq 2\}$ bounded?

222. Let M be a set of real $n \times n$ matrices such that (i) $I \in M$, where I is the $n \times n$ identity matrix; (ii) if $A \in M$ and $B \in M$ then either $AB \in M$ or $-AB \in M$ but not both; (iii) if $A \in M$ and $B \in M$ then either $AB = BA$ or $AB = -BA$; (iv) if $A \in M$ and $A \neq I$ then there is at least one $B \in M$ such that $AB = -BA$. Prove that M contains at most n^2 matrices.
223. Let A and B be different $n \times n$ matrices with real entries. If $A^3 = B^3$ and $A^2B = B^2A$, can $A^2 + B^2$ be invertible?
224. If A and B are square matrices of the same size such that $ABAB = O$, does it follow that $BABA = O$?
225. A transversal of an $n \times n$ matrix A consists of n entries of A , no two in the same row or column. Let $f(n)$ be the number of $n \times n$ matrices A satisfying the following two conditions: (a) each entry $a_{i,j}$ of A is in the set $\{-1, 0, 1\}$; (b) the sum of the n entries of a transversal is the same for all transversals of A . An example of such a matrix A is

$$A = \begin{pmatrix} -1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Determine with proof a formula for $f(n)$ of the form $f(n) = a_1b_1^n + a_2b_2^n + a_3b_3^n + a_4$ where the a_i and b_i are rational numbers.

226. Let G be a finite set of real $n \times n$ matrices $\{M_i\}$, $1 \leq i \leq r$, which form a group under matrix multiplication. Suppose that $\sum_{i=1}^r \text{tr}(M_i) = 0$ where $\text{tr}(A)$ denotes the trace of the matrix A . Prove that $\sum_{i=1}^r M_i$ is the $n \times n$ zero matrix.

227. Let $k \in \mathbb{N}^*$, $u_1, u_2, \dots, u_k \in \mathbb{C}^*$, $z_1, z_2, \dots, z_k \in \mathbb{C}^*$ be distinct, and let

$$a_n = u_1 z_1^n + u_2 z_2^n + \dots + u_k z_k^n, \quad n \in \mathbb{N}.$$

Show that if the set $\{a_n \mid n \in \mathbb{N}\}$ is finite, then there exists $d \in \mathbb{N}^*$ such that $a_n = a_{n+d}$ for every $n \in \mathbb{N}$.

228. Let $A \in M_n(\mathbb{R})$ with $\det A \neq 0$, and let $f_A : M_n(\mathbb{R}) \rightarrow \mathbb{R}$,

$$f_A(X) = \sum_{i,j=1}^n b_{ij}x_{ij}, \quad \text{where } B = AA^t.$$

Show that:

- (a) $f_A(XX^t) = 0 \Rightarrow X = 0$,
- (b) $[f_A(XY)]^2 \leq f_A(XX^t)f_A(Y^tY)$, for all $X, Y \in M_n(\mathbb{R})$.

229. Consider the function $f : M_{n,1}(\mathbb{R}) \rightarrow [0, \infty)$,

$$f(X) = \sum_{k=1}^n |x_k|,$$

for every $X \in M_{n,1}(\mathbb{R})$. Show that there is only a finite number of matrices $A \in M_n(\mathbb{R})$ with the property $f(AX) = f(X)$ for every $X \in M_{n,1}(\mathbb{R})$.

230. Consider the matrix

$$A_n = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{n+1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{1}{n} & \frac{1}{n+1} & \frac{1}{n+2} & \cdots & \frac{1}{2n-1} \end{pmatrix},$$

where n is a nonzero natural number.

- 1) Show that $\det A_n$ is nonzero.
- 2) Show that the sum of the entries of the inverse matrix A_n^{-1} equals n^2 .

231. Let $A, B \in M_n(\mathbb{C})$ and $\varepsilon = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$. Show that

$$\det(A+B) + \det(A+\varepsilon B) + \cdots + \det(A+\varepsilon^{n-1}B) = n[\det A + \det B].$$

232. Show that for every matrix $A \in M_n(\mathbb{C})$ there exist infinitely many natural numbers k such that the matrix $A^k + I_n$ is invertible.

233. Let $A, B \in M_n(\mathbb{C})$ have the property

$$|\det(A + zB)| \leq 1 \quad \text{for every } z \in \mathbb{C} \text{ with } |z| = 1.$$

Show that:

- (a) $|\det A + z \det B| \leq 1$ for every $z \in \mathbb{C}$ with $|z| = 1$.
- (b) If $A, B \in M_n(\mathbb{R})$, then $(\det A)^2 + (\det B)^2 \leq 1$.

234. Let $A \in M_n(\mathbb{R})$ be a matrix with positive entries satisfying

$$\sum_{k=1}^n a_{ik} = 1, \quad i = \overline{1, n}.$$

Show that A cannot have eigenvalues of modulus greater than 1.

235. Let the matrices $A, B, C \in M_n(\mathbb{C})$ be such that C commutes with A or with B . Show that

$$\det(AB + C) = \det(BA + C).$$

236. Let $A \in M_n(\mathbb{Q})$ be such that $\det(A - I_n) = 0$ and $A^p = I_n$, where p is a prime number. Show that $p - 1 \mid n$.

237. Let $S(X)$ denote the sum of all entries of the matrix X . Show that if $A \in M_n(\mathbb{C})$ and

- (a) $S(A) = S(A^2) = \dots = S(A^n) = n$, then $S(A^k) = n$ for every $k \in \mathbb{N}^*$.
- (b) $S(A) = n^2$, $S(A^2) = n^3$, $\dots$, $S(A^n) = n^{n+1}$, then $S(A^k) = n^{k+1}$ for every $k \in \mathbb{N}^*$.

238. Show that if there exists an invertible matrix $A \in M_n(\mathbb{Q})$ such that

$$A^{-1} = A^2 + A,$$

then n is divisible by 3.

239. Let $\sum_{n=0}^{\infty} a_n x^{2n}$ be the series expansion of the function $f(x) = \frac{1}{\cos x}$. The numbers $e_n = \frac{1}{(2n)!} a_n$ are called the *Euler numbers*. Show that $e_n = (2n)! D_n$, where

$$D_n = \begin{vmatrix} \frac{1}{2!} & 1 & 0 & 0 & \cdots & 0 \\ \frac{1}{4!} & \frac{1}{2!} & 1 & 0 & \cdots & 0 \\ \frac{1}{6!} & \frac{1}{4!} & \frac{1}{2!} & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{1}{(2n)!} & \frac{1}{(2n-2)!} & \frac{1}{(2n-4)!} & \frac{1}{(2n-6)!} & \cdots & \frac{1}{2!} \end{vmatrix}.$$

240. Let $n \geq 2$, $n \in \mathbb{N}$. Determine all polynomials $f \in \mathbb{C}[X]$ with the property

$$f(\text{Tr } A) = \text{Tr}(f(A)) \quad \text{for every } A \in M_n(\mathbb{C}).$$

241. Let $n \geq 2$, $n \in \mathbb{N}$. Determine all polynomials $f \in \mathbb{C}[X]$ with the property

$$\det(f(A)) = f(\det A) \quad \text{for every } A \in M_n(\mathbb{C}).$$

242. Determine the number of (m, n) -type matrices with entries $a_{ij} \in \{\pm 1\}$ for which the product of the entries of each row and of each column equals -1 .

243. Let $A \in M_3(\mathbb{C})$ with $\det A = 1$, $\text{Tr } A = \text{Tr } A^{-1} = 0$. Show that $A^3 = I_3$.

244. Let $A, B \in M_n(\mathbb{C})$ be such that $AB + A + B = 0$. Show that $AB = BA$.

245. Let $A, B \in M_n(\mathbb{R})$ be such that $A^2 + B^2 = AB$ and the matrix $AB - BA$ is invertible. Show that n is divisible by 3.

246. Let $A, B, C \in M_n(\mathbb{C})$ with A invertible and $(A - B)C = BA^{-1}$. Show that $C(A - B) = A^{-1}B$.

247. Let $A_1, A_2, \dots, A_k \in M_n(\mathbb{R})$ be such that

$$A_1 A_1^t + A_2 A_2^t + \cdots + A_n A_n^t = 0.$$

Show that $A_1 = A_2 = \cdots = A_n = 0$.

248. Let $A, B, C, D \in M_2(\mathbb{C})$ and denote $[A, B] = AB - BA$. Show that there exists $\lambda \in \mathbb{C}$ such that

$$[A, B] \cdot [C, D] - [C, D] \cdot [A, B] = \lambda I_2.$$

249. Let $A \in M_{3,2}(\mathbb{C})$, $B \in M_{2,3}(\mathbb{C})$ be such that

$$AB = \begin{pmatrix} 8 & 2 & -2 \\ 2 & 5 & 4 \\ -2 & 4 & 5 \end{pmatrix}.$$

Show that $BA = 9I_2$.

250. Let $A \in M_n(\mathbb{R})$, $A \neq 0$, be such that $a_{ik}a_{jk} = a_{kk}a_{ij}$ for all $i, j, k = \overline{1, n}$. Show that:

- (a) $\text{Tr } A \neq 0$.
- (b) $A^t = A$.
- (c) $f_A(x) = x^{n-1}(x - \text{Tr } A)$.

251. Let $A, B \in M_n(\mathbb{R})$ be such that $\text{Tr}(AA^t + BB^t) = \text{Tr}(AB + A^tB^t)$. Show that $A = B^t$.

252. Let $A, B \in M_n(\mathbb{C})$. Show that

$$\text{Tr}(AB) \leq \frac{1}{2} \text{Tr}(AA^* + BB^*).$$

253. Let $A \in M_n(\mathbb{C})$ be a nonsingular matrix with columns $A_1, A_2, \dots, A_n$, and let $B \in M_n(\mathbb{C})$ have columns $A_2, A_3, \dots, A_n, 0$. Show that the matrices $C = BA^{-1}$ and $D = A^{-1}B$ both have rank $n - 1$ and all of their eigenvalues equal to zero.

254. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ and a, b be real parameters. Consider the polynomial

$$p(x) = \det \begin{pmatrix} \lambda_1 + x & a + x & \cdots & a + x \\ b + x & \lambda_2 + x & \cdots & a + x \\ \vdots & \vdots & \ddots & \vdots \\ b + x & b + x & \cdots & \lambda_n + x \end{pmatrix}.$$

- (a) Prove that $p(x)$ is a polynomial of degree at most 1 with respect to x .

- (b) Calculate the determinant of order n when $x = 0$.
255. For a matrix $A \in M_n(\mathbb{R})$ and two column vectors $u, v \in \mathbb{R}^n$, consider the polynomial $p(x) = \det(A + x \cdot uv^T)$. Prove that $p(x)$ is a polynomial of degree ≤ 1 and find the coefficient of x .
256. Let $A, B \in M_n(\mathbb{R})$. Prove that if $\text{rank}(B) = k$, then the polynomial $p(x) = \det(A + xB)$ has a degree not exceeding k .
257. Let A be a real matrix of order n . Given that the sum of all elements in each row equals λ . Let J be the matrix of all ones. Calculate the determinant $\det(A - tI + J)$ in terms of the characteristic polynomial of A , denoted as $P_A(t) = \det(A - tI)$.
258. Let row matrices $A, B, C \in M_{1 \times n}(\mathbb{R})$.
- (a) Prove that if $Bx = 0 \Rightarrow Ax = 0$ for all $x \in \mathbb{R}^n$, then there exists a real number r such that $A = rB$.
 - (b) Prove that if $(Bx = 0 \text{ and } Cx = 0) \Rightarrow Ax = 0$ for all $x \in \mathbb{R}^n$, then there exist real numbers r, t such that $A = rB + tC$.
259. Let $f, g_1, g_2, \dots, g_k$ be linear functionals on a vector space V . Prove that $\bigcap_{i=1}^k \ker(g_i) \subseteq \ker(f)$ if and only if $f \in \text{span}\{g_1, g_2, \dots, g_k\}$.
260. Let $A \in M_{m \times n}(\mathbb{R})$ and $B \in M_{p \times n}(\mathbb{R})$. Prove that $\ker(A) \subseteq \ker(B)$ if and only if there exists a matrix $P \in M_{p \times m}(\mathbb{R})$ such that $B = P \cdot A$.
261. Let $f : M_n(\mathbb{R}) \rightarrow \mathbb{R}$ be a linear functional. Suppose $f(AB) = f(BA)$ for all matrices $A, B \in M_n(\mathbb{R})$. Prove that there exists a real number c such that $f(X) = c \cdot \text{tr}(X)$ for all $X \in M_n(\mathbb{R})$.
262. Let $A \in M_n(\mathbb{R})$ such that for all vectors $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, the two vectors xA and x are always linearly dependent in $\mathbb{R}^n$. Prove that there exists a real number a such that $A = aI_n$.

263. Let $A \in M_n(\mathbb{R})$ (with $n \geq 3$). Suppose that for every vector $x \in \mathbb{R}^n$, the set of three vectors $\{x, Ax, A^2x\}$ is always linearly dependent. Prove that the set $\{I, A, A^2\}$ is linearly dependent in the matrix space.
264. Let $A, B \in M_n(\mathbb{R})$ be matrices such that $\text{rank}(A) \geq 2$. Suppose that for every vector $x \in \mathbb{R}^n$, the vectors Ax and Bx are always linearly dependent. Prove that there exists a real constant k such that $B = kA$.
265. Let $A = (a_{ij})_{n \times n}$ be a matrix satisfying the strictly diagonally dominant condition: $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all $i = 1, \dots, n$. Prove that A is invertible.
266. Let A be a real skew-symmetric matrix of order n .
- (a) If n is odd, prove that A is not invertible.
 - (b) Let $M = I_n + A$. Prove that M is always invertible.
 - (c) Prove that $\det(I + A) \geq 1$ for all real skew-symmetric matrices A .
 - (d) Prove that the matrix $Q = (I - A)(I + A)^{-1}$ always has a determinant of 1 and is an orthogonal matrix.
267. Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be pairwise distinct real numbers such that $a_i + b_j \neq 0$. Prove that the matrix C with elements $c_{ij} = \frac{1}{a_i + b_j}$ is invertible.
268. Let A satisfy $A^k = 0$ for some positive integer k (nilpotent matrix). Prove that $I - A$ is always invertible.
269. Let A be a square matrix of order n satisfying: every main diagonal element a_{ii} is even, and every off-diagonal element a_{ij} ($i \neq j$) is odd. Prove that if n is even, A is invertible over $\mathbb{R}$.
270. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ be a block matrix where A and D are square matrices. Assume A is invertible. Prove that M is invertible if and only if the Schur complement $S = D - CA^{-1}B$ is invertible.
271. Let A be an $n \times n$ matrix with elements $a_{ij} = \gcd(i, j)$. Prove that A is invertible.

272. Let $x_1 < x_2 < \cdots < x_n$ be distinct real numbers. Consider matrix A with $a_{ij} = |x_i - x_j|$. Prove that for $n > 1$, A is always invertible.
273. Let C be a circulant matrix with first row $(c_0, c_1, \dots, c_{n-1})$. Prove C is invertible if and only if $f(\omega^k) \neq 0$ for all $k = 0, \dots, n-1$, where $f(x) = c_0 + c_1x + \cdots + c_{n-1}x^{n-1}$ and $\omega = e^{2\pi i/n}$.
274. Let A and B be two real symmetric positive definite matrices. Prove that their Hadamard product $C = A \circ B$ (where $c_{ij} = a_{ij}b_{ij}$) is also positive definite and thus invertible.
275. (Sylvester's Rank Inequality) For an $m \times n$ matrix A and an $n \times p$ matrix B , prove that
- $$\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B)).$$
276. (Frobenius Rank Inequality) For matrices A, B, C such that the product ABC exists, prove that
- $$\text{rank}(AB) + \text{rank}(BC) \leq \text{rank}(B) + \text{rank}(ABC).$$
277. Let P be a square matrix satisfying $P^2 = P$ (a projection matrix).
- (a) Prove that $\text{rank}(P) + \text{rank}(I - P) = n$.
 - (b) Prove that $\text{rank}(P) = \text{tr}(P)$.
278. Prove that any $m \times n$ matrix A of rank r can be factored as $A = B \cdot C$, where B is an $m \times r$ matrix (full column rank) and C is an $r \times n$ matrix (full row rank).
279. Let A be a square matrix and u, v be column vectors. Prove that $\text{rank}(A + uv^T)$ can only take the values: $\text{rank}(A) - 1$, $\text{rank}(A)$, or $\text{rank}(A) + 1$.
280. Let A, B be $n \times n$ matrices. Prove that $\text{rank}(A + B) = \text{rank}(A) + \text{rank}(B)$ if and only if $\text{Im}(A) \cap \text{Im}(B) = \{0\}$ and $\text{Im}(A^T) \cap \text{Im}(B^T) = \{0\}$.
281. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Prove that $\text{rank}(AB) = \text{rank}(B) - \dim(\text{Im}(B) \cap \ker(A))$. Deduce the necessary and sufficient condition for $\text{rank}(AB) = \text{rank}(B)$.

282. Let A be a real matrix satisfying $A^3 = A$. Prove that $\text{rank}(A) = \text{tr}(A^2)$.

283. Calculate the determinant of order n :

$$D = \det \begin{pmatrix} 1+x_1 & 1 & 1 & \cdots & 1 \\ 1 & 1+x_2 & 1 & \cdots & 1 \\ 1 & 1 & 1+x_3 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1+x_n \end{pmatrix}.$$

284. Calculate the determinant of order n ($n \geq 3$) with elements $a_{ij} = \cos(x_i + y_j)$.

285. Let $S = \{x_1, \dots, x_n\}$ be a set of n distinct positive integers, closed under the GCD operation. Prove that $\det(\gcd(x_i, x_j)) = \prod_{i=1}^n \varphi(x_i)$.

286. Let A and B be square matrices of order n . If $\text{rank}(B) = 1$, prove that the function $f(x) = \det(A + xB)$ is a polynomial of at most degree 1 in x .

287. Let A be an $n \times n$ matrix with $A_{ii} = \alpha$ and $A_{ij} = \beta$ for $i \neq j$. Calculate $\det(A)$.

288. Let $f : V \rightarrow V$ be an endomorphism on a finite-dimensional vector space V .

(a) Prove there exists a positive integer k such that $\ker(f^k) = \ker(f^{k+1})$ and $\text{Im}(f^k) = \text{Im}(f^{k+1})$.

(b) With such k , prove that $V = \ker(f^k) \oplus \text{Im}(f^k)$.

289. Let $f : V \rightarrow W$ and $g : V \rightarrow U$ be linear mappings. Prove there exists a linear mapping $h : W \rightarrow U$ such that $g = h \circ f$ if and only if $\ker(f) \subseteq \ker(g)$.

290. Let $f : V \rightarrow V$ satisfy $f^2 = 0$. Prove that $\text{Im}(f) \subseteq \ker(f)$. Furthermore, for an arbitrary f , prove that $\dim(\text{Im}(f) \cap \ker(f)) = \dim(\text{Im}(f)) - \dim(\text{Im}(f^2))$.

291. Let V be an n -dimensional vector space. Let $H_1, H_2, \dots, H_k$ be k hyperplanes (subspaces of dimension $n-1$). Prove that $\dim(H_1 \cap H_2 \cap \cdots \cap H_k) \geq n - k$.

292. Let $f : U \rightarrow V$ and $g : V \rightarrow W$ be linear mappings. Prove the inequality: $\dim(\ker(g \circ f)) \leq \dim(\ker(f)) + \dim(\ker(g))$.
293. Let A, B be square matrices of order n over $\mathbb{C}$.
- (a) Prove that the equality $AB - BA = I$ is impossible.
 - (b) Suppose $AB - BA = A$. Prove that A is nilpotent.
294. Let P, Q be two projections ($P^2 = P, Q^2 = Q$). Prove that $\text{tr}(P - Q)^3 = \text{tr}(P - Q)$.
295. Let $f : V \rightarrow V$ be an endomorphism. Prove that $\text{tr}(f) = 0$ if and only if there exist two endomorphisms $g, h : V \rightarrow V$ such that $f = g \circ h - h \circ g$.
296. Let A and B be diagonalizable matrices of order n . Prove that A and B commute ($AB = BA$) if and only if they are simultaneously diagonalizable.
297. Let A be a matrix over $\mathbb{C}$ with n distinct eigenvalues. Prove that a matrix B commutes with A if and only if B is a polynomial in A .
298. Let A be an $n \times n$ matrix over $\mathbb{C}$. Consider the linear mapping $\phi_A(X) = AX - XA$. Prove that ϕ_A is diagonalizable if and only if A is diagonalizable.
299. Let $A \in M_n(\mathbb{Q})$ satisfy $A^p = I$ where p is prime. Prove that A is diagonalizable over $\mathbb{Q}$ if and only if $A = I$.
300. Prove that a complex matrix A is diagonalizable if and only if for all $\lambda \in \mathbb{C}$, $\text{rank}(A - \lambda I) = \text{rank}((A - \lambda I)^2)$.
301. Let A be a complex normal matrix ($AA^* = A^*A$). Prove it is unitarily diagonalizable.
302. Let A be a complex matrix of order n . Prove that A is nilpotent ($A^n = 0$) if and only if $\text{tr}(A^k) = 0$ for all $k = 1, 2, \dots, n$.
303. The Contrast Matrix of order $n \geq 2$, denoted K_n , is defined by $K_{ij} = \frac{i}{j} - \frac{j}{i}$.
- (a) Prove K_n is skew-symmetric with trace 0.

- (b) Prove $\text{rank}(K_n) = 2$.
- (c) Prove the characteristic polynomial has the form $P_K(\lambda) = \lambda^{n-2}(\lambda^2 + \Delta_n)$.
304. Let $S(n, m)$ be the set of matrices where the sum of even-indexed columns equals the sum of odd-indexed columns. If $A \in S(n, m)$, find the smallest eigenvalue of the matrix $M = I + A^T A$.
305. Let A be an $n \times m$ real matrix with $n < m$. Prove that $\lambda = 1$ is an eigenvalue of $M = I_m + A^T A$ with algebraic multiplicity at least $m - n$.
306. Let $A \in M_{2 \times 2}(\mathbb{C})$ have two distinct eigenvalues. Let $S = \left\{ N \in M_{2 \times 2}(\mathbb{C}) \mid \begin{pmatrix} A & N \\ 0 & A \end{pmatrix} \text{ is diagonalizable} \right\}$. Prove that S is a 2-dimensional subspace of $M_{2 \times 2}(\mathbb{C})$.
307. Let $a_1, \dots, a_n \in \mathbb{C} \setminus \{0\}$ and $A = \left(\frac{a_i}{a_j} \right)_{1 \leq i, j \leq n}$. Prove A is diagonalizable and find an invertible matrix C such that $C^{-1}AC$ is diagonal.
308. Does there exist a matrix $A \in M_{3 \times 3}(\mathbb{R})$ such that $\text{tr}(A) = 0$ and $A^2 + A^T = I$? Prove your assertion.
309. Let $A = (a_{ij}) \in M_{n \times n}(\mathbb{R})$ satisfy $a_{ii} = 0$ for all i and $a_{ij} > 0$ for all $i \neq j$. Find the minimum possible value of $\text{rank}(A)$.
310. Let V be a k -dimensional real vector space with standard inner product. Given a system of n vectors $x_1, x_2, \dots, x_n \in V$, consider the matrix $D \in M_n(\mathbb{R})$ defined by the squared distances $D_{ij} = \|x_i - x_j\|^2$. Prove that $\text{rank}(D) \leq k + 2$.
311. Calculate the following determinant of order n :

$$A_n = \begin{pmatrix} 2021 & 1 & 1 & \cdots & 1 & 1 \\ 1 & 2021 & 1 & \cdots & 1 & 1 \\ 1 & 1 & 2021 & \cdots & 1 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & 2021 & 1 \\ 1 & 1 & 1 & \cdots & 1 & 2021 \end{pmatrix}.$$

312. Calculate the following determinant of order n :

$$B_n = \begin{pmatrix} 5 & 3 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 2 & 5 & 3 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 2 & 5 & 3 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 5 & 3 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 2 & 5 \end{pmatrix}.$$

313. Calculate the following determinant of order n , where $a, b \in \mathbb{R}$:

$$C_n = \begin{pmatrix} a+b & ab & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & a+b & ab & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & a+b & ab & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a+b & ab & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & a+b & ab \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & a+b \end{pmatrix}.$$

314. Let M be an $n \times n$ matrix where every diagonal entry is a and every off-diagonal entry is b :

$$M = \begin{pmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \cdots & a \end{pmatrix}.$$

Show that

$$M^{-1} = \frac{1}{a-b}I - \frac{b}{(a-b)(a+(n-1)b)}J$$